



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Compactness in Metric Spaces

Heine-Borel Theorem and Extreme Value Theorem

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CONDUCTED BY

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Open Cover and Subcover

Open Cover

Let A be a subset of a metric space (X, d) . An *open cover* of A is a collection $\{U_\alpha\}_{\alpha \in I}$ of open subsets of X such that

$$A \subseteq \bigcup_{\alpha \in I} U_\alpha.$$

Subcover

A *subcover* of an open cover $\{U_\alpha\}_{\alpha \in I}$ of a set A is a subcollection $\{U_\alpha\}_{\alpha \in J}$, where $J \subseteq I$, such that

$$A \subseteq \bigcup_{\alpha \in J} U_\alpha.$$

Finite Subcover

A *finite subcover* of an open cover $\{U_\alpha\}_{\alpha \in I}$ of a set A is a subcollection $\{U_\alpha\}_{\alpha \in J}$, where $J \subseteq I$ is a finite set, such that

$$A \subseteq \bigcup_{\alpha \in J} U_\alpha.$$

Compactness

Compactness

A subset A of a metric space (X, d) is called *compact* if every open cover of A has a finite subcover. That is, if $\{U_\alpha\}_{\alpha \in I}$ is an open cover of A , then there exists a finite subset $J \subseteq I$ such that

$$A \subseteq \bigcup_{\alpha \in J} U_\alpha.$$

② Problem

Show that the Real line $\mathbb{R}$ is not compact in the standard metric.



Problem

Show that the closed interval $(0, 1)$ is not compact in the standard metric on $\mathbb{R}$.

Problem

Show that any finite subset of real numbers is compact in the standard metric on $\mathbb{R}$.

❓ Problem

Show that the closed interval $[0, 1]$ is compact in the standard metric on $\mathbb{R}$.

Why Compactness is Important?

Boundedness of Functions

💡 Boundedness of Continuous Functions on Compact Sets

Let (X, d) be a metric space and let $f : X \rightarrow \mathbb{R}$ be a continuous function. If $K \subseteq X$ is compact, then f is bounded.

Some Important Theorems on Compactness

Theorem

Every closed subset of a compact metric space is compact.



Theorem

Every compact subset of a metric space (X, d) is closed.



Theorem

Every compact subset of a metric space (X, d) is bounded.



Heine-Borel Theorem

Heine-Borel Theorem

A subset of $\mathbb{R}$ is compact if and only if it is closed and bounded.



Continuous Image and Extreme Value Theorem

💡 Continuous Image of Compact Set is Compact

Let (X, d) be a metric space and let $f : X \rightarrow \mathbb{R}$ be a continuous function. If $K \subseteq X$ is compact, then $f(K)$ is compact in $\mathbb{R}$.

Extreme Value Theorem

💡 Extreme Value Theorem

Let (X, d) be a metric space and let $f : X \rightarrow \mathbb{R}$ be a continuous function. If $K \subseteq X$ is compact, then f attains its maximum and minimum values on K .

Thank You!

We'd love your questions and feedback.

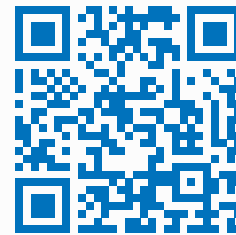
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(Lectures, walkthroughs, and course updates)



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References

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- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.